

NV23 image172647 closeout report

Aligned NV, T2*, and 13C evidence | OpenClaw nv-researcher | 2026-05-15

Objective: find a magnetic-field-aligned NV from image172647, then obtain well-supported T2* and 13C conclusions. Final branch: reimage1804_c02. Strong-pi pODMR showed a clear signal-only resonance near 3.87646 GHz. Ramsey data at detunings 1.50, 1.00, and 1.25 MHz support a short/few-microsecond T2* scale, not a precise scalar. 13C conclusion: likely weak/moderate nearby-13C-like coupling. Ramsey alone was not claim-grade, but terminal CPMG N=8 independently showed a final-echo dip in the modeled 1/(4 f13) target region.

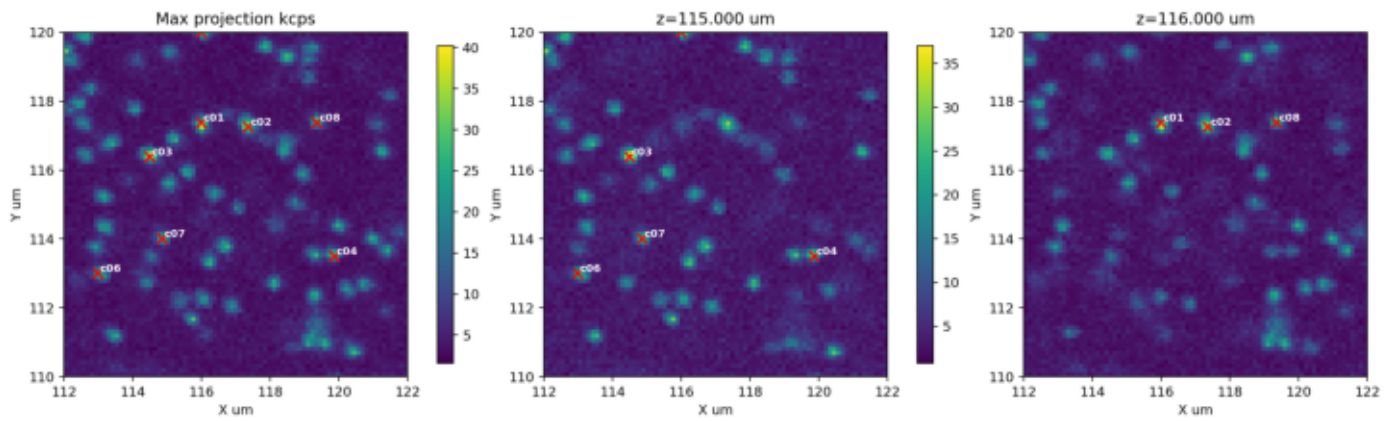
Final claims

- Aligned NV established: reimage1804_c02 tracked at 39.690 kcps and passed terminal strong-pi pODMR. Raw/fitted-reference pODMR depth was 13.8--14.0%, center 3.876461010 GHz, FWHM 11.2 MHz, no matching reference dip.
- T2*: report as short/few-us order. Fits are sensitive to early-tau handling: all-point fits about 1.5--2.1 us; excluding the first point gives about 2.4--3.1 us.
- 13C: likely weak/moderate nearby-13C-like coupling. CPMG target-region minimum at tau=0.6736 us (23.6 ns from modeled target) has readout3/self=0.9528, readout3/readout1=0.9518, and readout3/readout2=0.9568.
- No further bridge-touching work is required for the original objective. Optional refinement only if requested: fine CPMG around 0.60--0.72 us or phase-readout XY8/DDRF-style discriminator.

Candidate search and pODMR acceptance

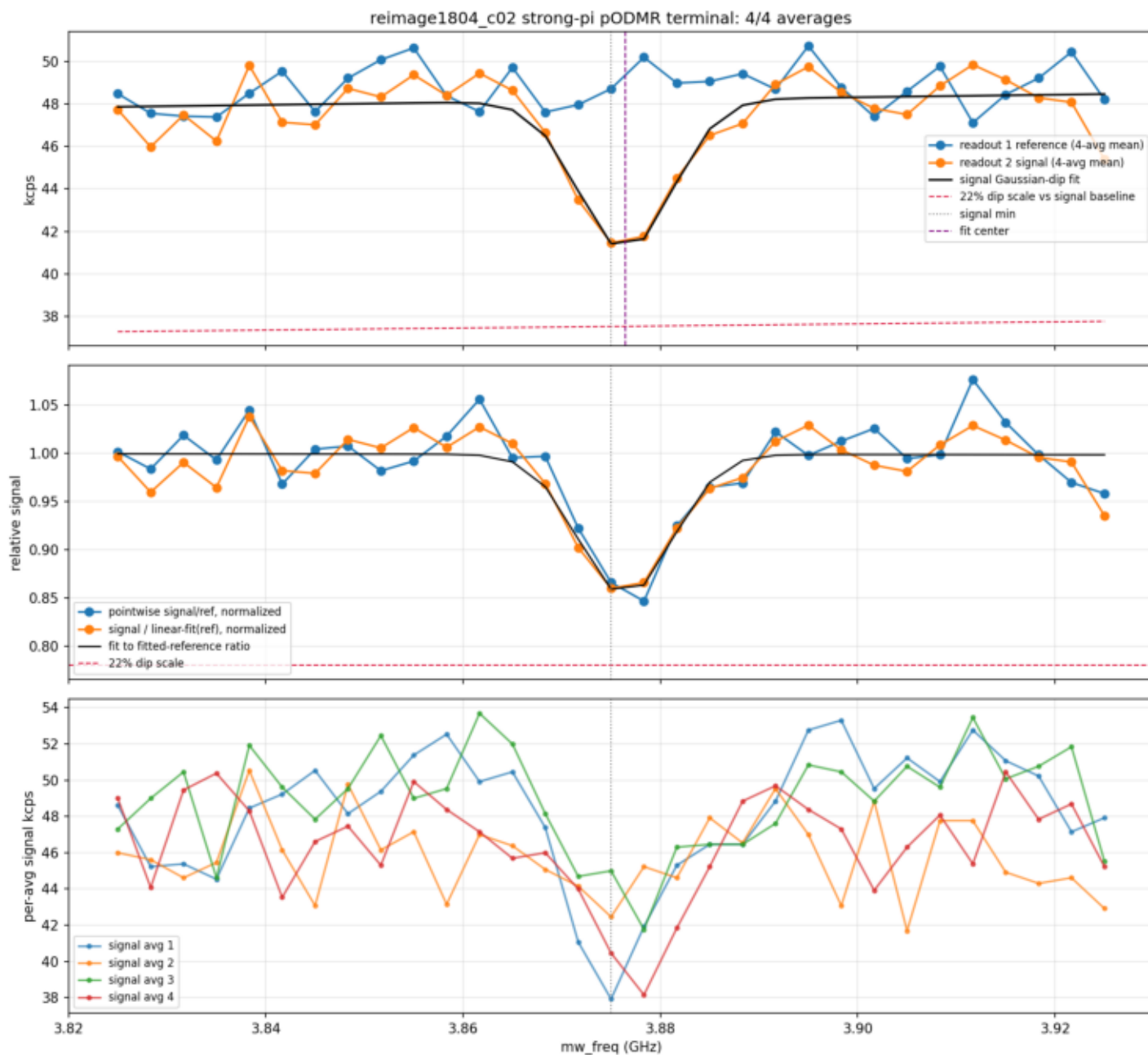
Initial image172647_c01 tracked once, but the first pODMR execute failed before data with low in-run counts and immediate standalone retrack also failed low-count. This was treated as freshness/count evidence, not no-resonance evidence. Fresh re-imaging produced candidates reimage1804_c01, c02, and c03. reimage1804_c01 tracked but terminal strong-pi pODMR showed only about 4--5% depressions versus expected ~22%, so it was rejected. reimage1804_c02 then tracked successfully and passed strong-pi pODMR alignment screening. Weak-pi pODMR refined the microwave center to 3.876501337 GHz with depth about 11.4% and FWHM about 0.94 MHz; this center was used for the det-shift Ramsey and CPMG follow-up.

Fresh candidate ranking



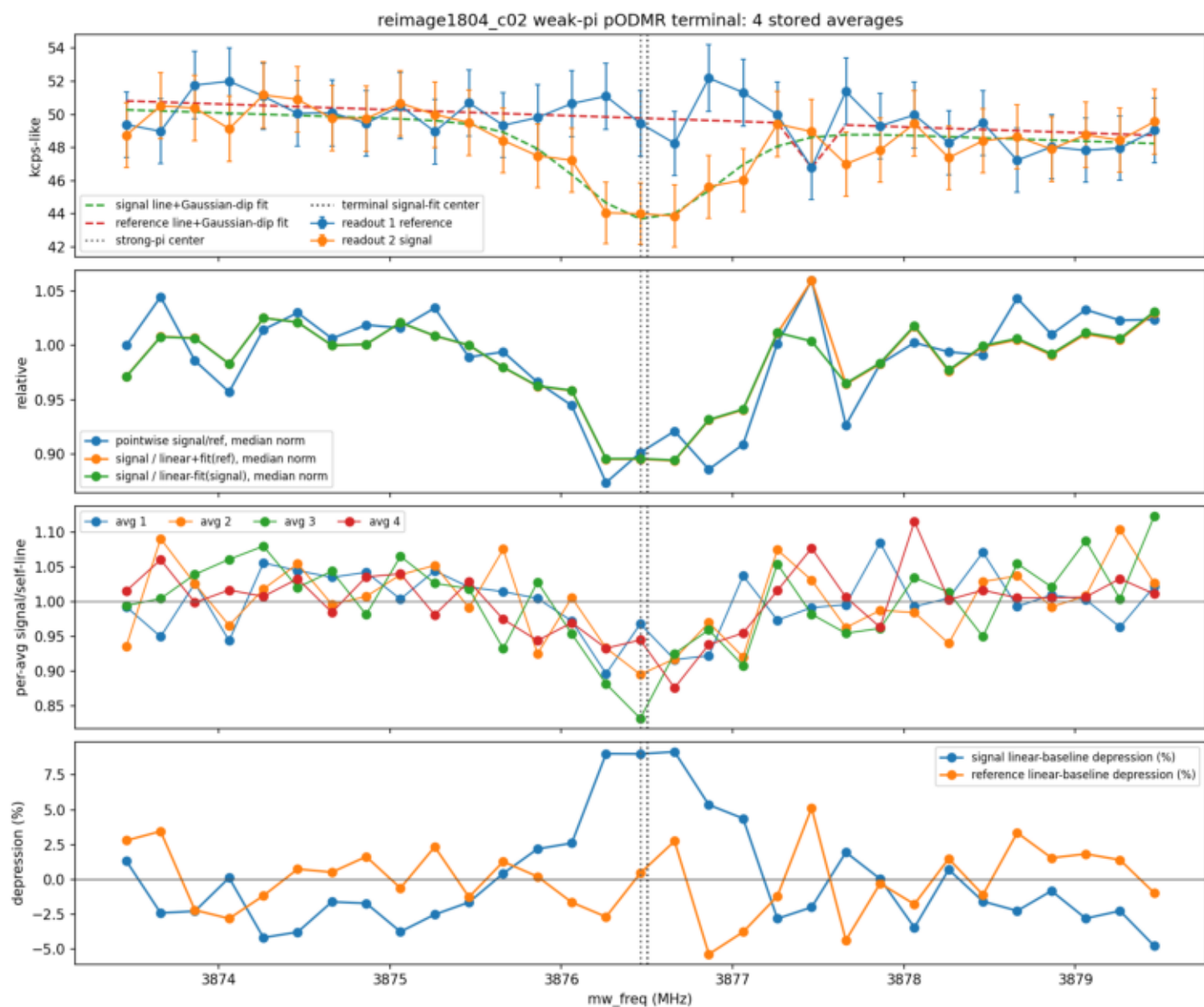
Fresh re-image candidate ranking after the stale initial candidate failed low-count diagnostics.

Strong-pi pODMR: accepted aligned branch



Terminal strong-pi pODMR for reimage1804_c02: signal-only dip near 3.87646 GHz with lower-than-reference but usable contrast.

Weak-pi pODMR refinement



Terminal weak-pi pODMR refinement: center close to strong-pi center; supports using 3.876501337 GHz for later Ramsey/CPMG planning.

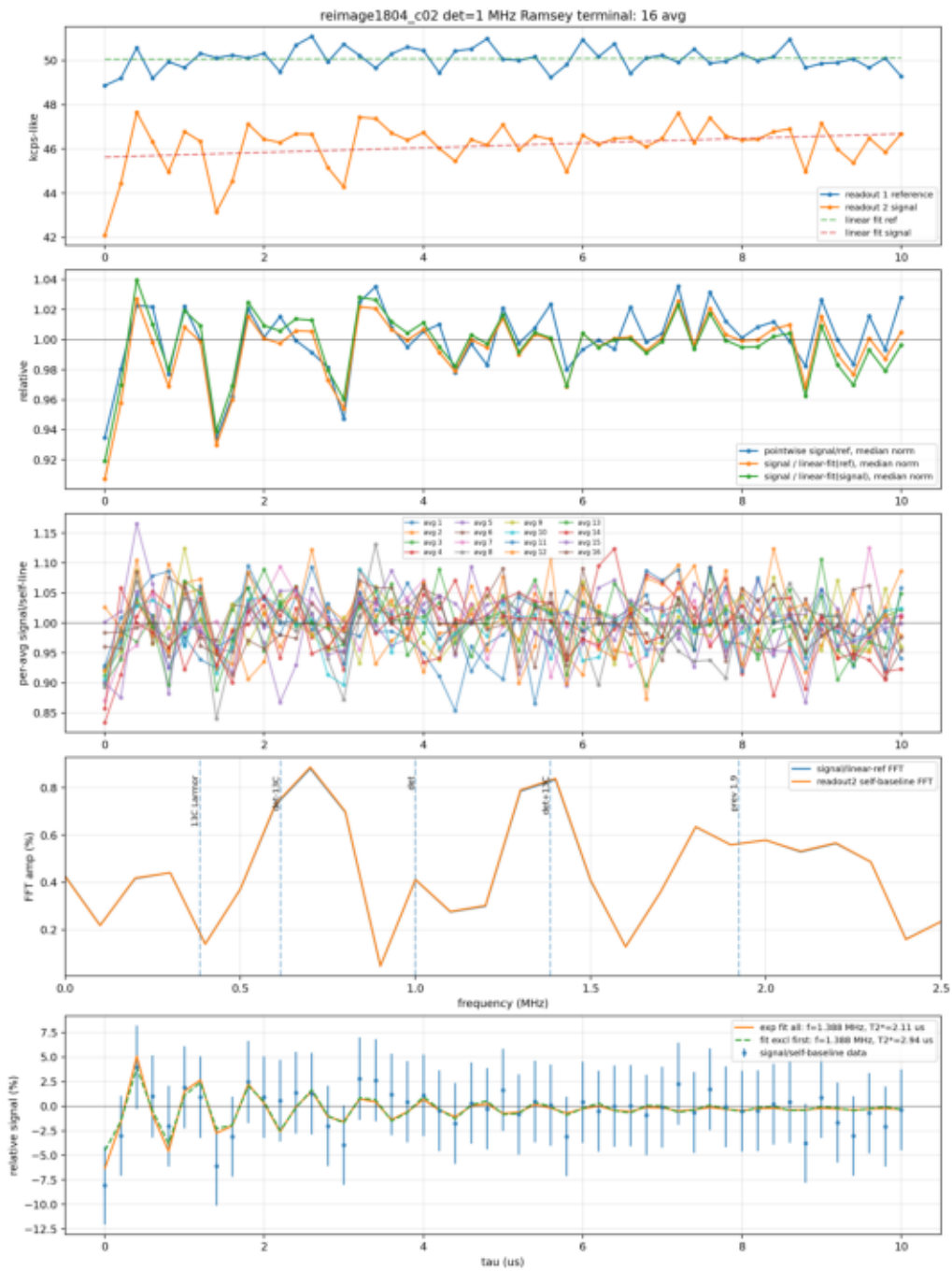
Ramsey evidence summary

Terminal Ramsey target amplitudes (readout2/self FFT, %) and fit-derived T2* scale

det MHz	shots	span us	direct13C	det-13C	carrier	det+13C	T2 all us	T2 excl1 us
1.50	8x50k	0-8	1.39	0.79	0.45	1.95	1.47	2.39
1.00	16x50k	0-10	0.14	0.73	0.41	0.84	2.11	2.94
1.25	16x50k	0-10	0.54	0.79	1.00	1.01	2.12	3.08

Interpretation: high-sideband target amplitudes are approximately 1.95%, 0.84%, and 1.01% for det=1.50, 1.00, and 1.25 MHz, so a weak det-shift-consistent component remains plausible. However, low-sideband/direct-Larmor/carrier support is incomplete and all features are about-percent or below, far below the pODMR resonance contrast. Ramsey alone is not a claim-grade 13C observation. T2* is short/few-us order but not a precise scalar.

Terminal det=1.0 MHz Ramsey



Terminal det=1.0 MHz Ramsey: short/few-us T2* order and weak high-sideband candidate, but no standalone 13C claim.

Terminal det=1.25 MHz Ramsey



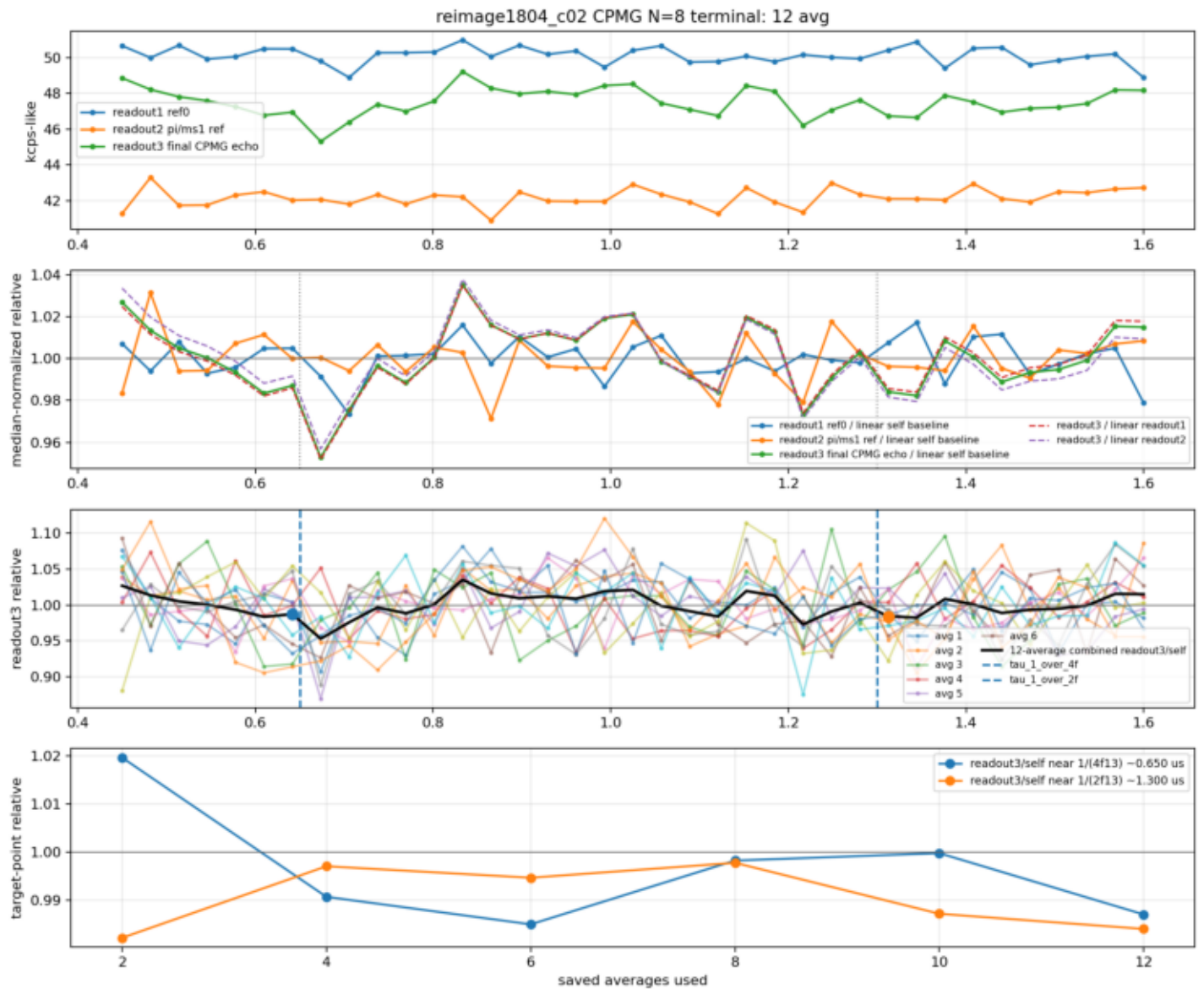
Terminal det=1.25 MHz Ramsey: old static high/1.9-MHz alternatives are weak; target features remain low contrast.

CPMG N=8 target-region corroboration

CPMG model: f13 about 384.6 kHz from the weak-pi pODMR center. Target conventions are $1/(4 f13)=0.6500$ us and $1/(2 f13)=1.3001$ us. Scan: CPMG.xml, N=8, tau=0.45--1.60 us, 37 points, 12x50k reps, 600k shots/point. Clean terminal bridge state: pre-run 38.694 kcps, post-run 44.535 kcps, no drift flags, no stop request. The target-region minimum at 0.6736 us is inside the modeled width scale and appears in readout3 self-baseline plus both reference-normalized views.

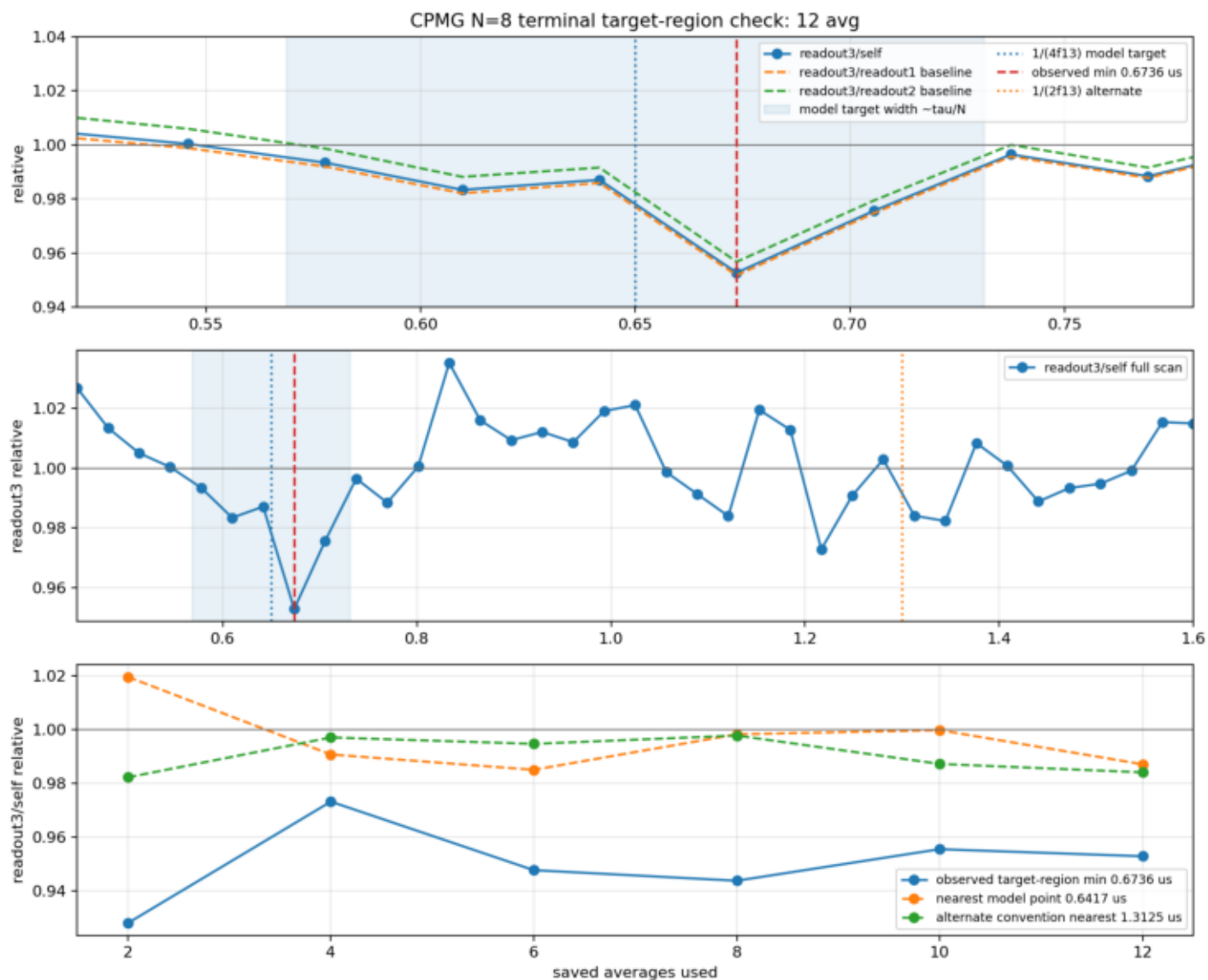
avg	shots/pt	r3/self @0.6736	dip %	nearest 0.650	nearest 1.300
2	100k	0.9279	7.21	1.0197	0.9822
4	200k	0.9732	2.68	0.9908	0.9971
6	300k	0.9476	5.24	0.9850	0.9947
8	400k	0.9437	5.63	0.9983	0.9978
10	500k	0.9555	4.45	0.9998	0.9872
12	600k	0.9528	4.72	0.9871	0.9841

Terminal CPMG N=8 full scan



Terminal 12-average CPMG N=8 full scan over tau=0.45--1.60 us.

Terminal CPMG target-region check



Target-region review: the 0.6736 us minimum is within the modeled 1/(4 f13) region and corroborates the weak Ramsey 13C candidate.

Limitations, references, and artifacts

Limitations: pODMR contrast is lower than the setup reference; T_2^* scalar is fit-window/early-time sensitive; ^{13}C evidence is likely weak/moderate and not sufficient for precise coupling extraction or publication-grade single-spin assignment. Optional follow-up, only if stronger confidence is requested, is a fine CPMG scan around 0.60--0.72 μs or a phase-readout XY8/DDRF-style discriminator after fresh model/advisory/verifier gates.

Literature/context check: local literature index includes Doherty et al., "The nitrogen-vacancy colour centre in diamond," *Physics Reports* 528, 1--45 (2013), doi:10.1016/j.physrep.2013.02.001. Crossref was checked for de Lange et al., "Universal dynamical decoupling of a single solid-state spin from a spin bath," *Science* 330, issue 6000 (2010), doi:10.1126/science.1192739. These references provide context only; final numeric claims here come from project artifacts.

Key artifact bundle: work/reports/closeout_20260515_1225/ (this PDF, LaTeX source, and copied figures). Primary evidence JSONs are the terminal pODMR, weak- π pODMR, Ramsey $\det=1.5/\det=1.0/\det=1.25$ summaries, and CPMG terminal/target-region summaries under work/artifacts/analysis/.